Light

GCSE Computer Science — Data Representation: Media (Set 1)

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GCSE • Edexcel • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

00:00 Start Pause Reset

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Multiple-choice topic test for **GCSE Computer Science (Edexcel)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

ASCII is a standard method for encoding text characters.
How many bits does standard ASCII use to represent each character?

- A. 5 bits
- B. 16 bits
- C. 8 bits
- D. 7 bits

Q2 (1 mark)

How many different characters can be represented using 7-bit ASCII?

- A. 64

- B. 256
- C. 512
- D. 128

Q3 (1 mark)

A bitmap image is made up of small squares of colour.
What is each small square called?

- A. A pixel
- B. A byte
- C. A vector
- D. A sample

Q4 (1 mark)

The resolution of an image refers to:

- A. The file size of the image
- B. The number of colours used
- C. The compression method used
- D. The number of pixels in the image

Q5 (1 mark)

Colour depth refers to:

- ○ **A.** The number of pixels in an image
- ○ **B.** The number of bits used to store each pixel's colour
- ○ **C.** The physical size of the image
- ○ **D.** The brightness of the image

Q6 (1 mark)

Sound waves are analogue signals.
What must happen to an analogue sound wave before a computer can store it?

- ○ **A.** It must be compressed
- ○ **B.** It must be converted to digital data
- ○ **C.** It must be amplified
- ○ **D.** It must be filtered

Q7 (1 mark)

Sample rate measures:

- ○ **A.** How many times per second the sound wave is measured
- ○ **B.** The volume of the sound
- ○ **C.** The number of bits used to store each sample
- ○ **D.** The length of the audio file

Q8 (1 mark)

Bit depth in digital audio refers to:

- ○ **A.** The number of samples taken per second
- ○ **B.** The length of the audio recording
- ○ **C.** The frequency range of the sound
- ○ **D.** The number of bits used to store each sample

Q9 (1 mark)

An image uses 1-bit colour depth.
How many different colours can each pixel be?

- ○ **A.** 2
- ○ **B.** 1
- ○ **C.** 8
- ○ **D.** 256

Q10 (1 mark)

Which of the following is included in the 7-bit ASCII character set?

- ○ **A.** Chinese characters
- ○ **B.** Emoji symbols
- ○ **C.** Upper and lowercase English letters

- ○ **D.** Mathematical symbols like π

Q11 (1 mark)

Increasing the sample rate of a digital audio recording will:

- ○ **A.** Decrease the file size
- ○ **B.** Reduce the audio quality
- ○ **C.** Make the recording shorter
- ○ **D.** Improve the audio quality

Q12 (1 mark)

Increasing the resolution of a bitmap image will:

- ○ **A.** Decrease the file size
- ○ **B.** Reduce the image quality
- ○ **C.** Increase the file size
- ○ **D.** Change the colour depth

Q13 (2 marks)

In ASCII, the character 'A' has the denary code 65.
What is the ASCII code for 'D'?

- ○ **A.** 66

- B. 67
- C. 68
- D. 69

Q14 (2 marks)

In ASCII, uppercase 'A' is 65 and lowercase 'a' is 97.
What is the ASCII code for lowercase 'f'?

- A. 70
- B. 100
- C. 102
- D. 134

Q15 (2 marks)

An image uses 8-bit colour depth.
How many different colours can be displayed?

- A. 8
- B. 64
- C. 256
- D. 65536

Q16 (2 marks)

A black and white image (1-bit colour) has dimensions of 800 × 600 pixels.
What is the file size in bits?

- ○ **A.** 480,000 bits
- ○ **B.** 960,000 bits
- ○ **C.** 3,840,000 bits
- ○ **D.** 4,800,000 bits

Q17 (2 marks)

An audio file is recorded at 44,100 Hz sample rate with 16-bit depth for 1 second.
What is the file size?

- ○ **A.** 44,100 bits
- ○ **B.** 352,800 bits
- ○ **C.** 705,600 bits
- ○ **D.** 1,411,200 bits

Q18 (2 marks)

A sound is sampled at 8,000 Hz.
What is the sample interval (time between samples)?

- ○ **A.** 0.0000125 seconds
- ○ **B.** 0.00125 seconds
- ○ **C.** 0.0125 seconds

- ○ **D.** 0.000125 seconds

Q19 (2 marks)

An image has dimensions 1024 × 768 pixels and uses 24-bit colour depth.
What is the file size in bytes?

- ○ **A.** 786,432 bytes
- ○ **B.** 2,359,296 bytes
- ○ **C.** 6,291,456 bytes
- ○ **D.** 18,874,368 bytes

Q20 (2 marks)

A digital photograph uses 8-bit colour depth but the original scene contains thousands of subtle colour variations.
What limitation occurs?

- ○ **A.** The image will be too large to store
- ○ **B.** Some colours will be approximated or lost
- ○ **C.** The image resolution will decrease
- ○ **D.** The file will become corrupted

Q21 (2 marks)

Which combination of settings will produce the highest quality digital audio?

- ○ **A.** 8,000 Hz sample rate, 8-bit depth
- ○ **B.** 22,050 Hz sample rate, 8-bit depth
- ○ **C.** 44,100 Hz sample rate, 16-bit depth
- ○ **D.** 22,050 Hz sample rate, 16-bit depth

Q22 (2 marks)

The ASCII code for 'M' is 77 in denary.
What is this in binary?

- ○ **A.** 1001011
- ○ **B.** 1001101
- ○ **C.** 1001111
- ○ **D.** 1010001

Q23 (2 marks)

A sound recording uses a low sample rate of 8,000 Hz.
What is a likely limitation of this recording?

- ○ **A.** The file will be too large to transmit
- ○ **B.** High-frequency sounds may not be captured accurately
- ○ **C.** The recording will be too quiet
- ○ **D.** The file format will be incompatible

Q24 (3 marks)

A digital camera takes photos at 4000 × 3000 pixels with 24-bit colour. How many photos can fit on a 4 GiB memory card?
(1 GiB = 1024^3 bytes)

- ○ **A.** Approximately 119 photos
- ○ **B.** Approximately 143 photos
- ○ **C.** Approximately 238 photos
- ○ **D.** Approximately 286 photos

Q25 (3 marks)

A stereo audio file is recorded at 44,100 Hz sample rate with 16-bit depth for 3 minutes.
What is the approximate file size in MiB?
(Stereo = 2 channels, 1 MiB = 1024^2 bytes)

- ○ **A.** Approximately 15 MiB
- ○ **B.** Approximately 30 MiB
- ○ **C.** Approximately 45 MiB
- ○ **D.** Approximately 60 MiB

Q26 (3 marks)

Image A is 640 × 480 with 24-bit colour. Image B is 1280 × 960 with 8-bit colour.
Which statement correctly compares their file sizes?

- ○ **A.** Image A is larger because it has more colours

- ○ **B.** Image B is larger because it has more pixels
- ○ **C.** They are the same size
- ○ **D.** Image A is larger because 24 > 8

Q27 (3 marks)

The word 'Help' is stored using ASCII.
If 'H' = 72, what is the total of all ASCII codes for 'Help'?

- ○ **A.** 393
- ○ **B.** 395
- ○ **C.** 397
- ○ **D.** 399

Q28 (3 marks)

Recording A uses 22,050 Hz sample rate and 8-bit depth. Recording B uses 11,025 Hz sample rate and 16-bit depth.
Both recordings are 2 minutes long. Which statement is correct?

- ○ **A.** Recording A is larger than Recording B
- ○ **B.** Recording B is larger than Recording A
- ○ **C.** Both recordings are exactly the same size
- ○ **D.** Recording A has better quality than Recording B

Q29 (3 marks)

A sensor records temperature readings as 8-bit unsigned values, where 0 represents -20°C and 255 represents 107°C.
What is the precision (smallest detectable change) of this system?

- ○ **A.** 0.25°C
- ○ **B.** 0.5°C
- ○ **C.** 1°C
- ○ **D.** 2°C

Q30 (3 marks)

A web designer must reduce an image file from 2 MiB to under 500 KiB without changing dimensions.
The image is currently 1600 × 1200 pixels with 32-bit colour.
What colour depth should be used?

- ○ **A.** 24-bit colour
- ○ **B.** 16-bit colour
- ○ **C.** 8-bit colour
- ○ **D.** 4-bit colour

Total: **55** marks • Time: **30 min**

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